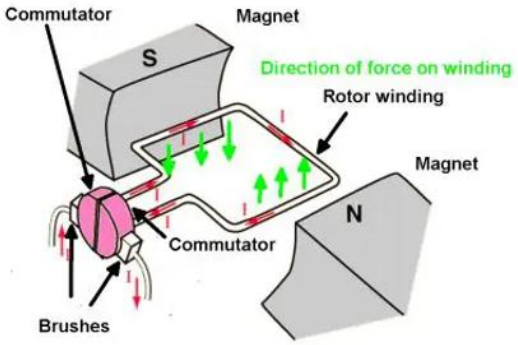
Chapter 50 – Electric Motors, Generators and Power Transmission
Calculus Physics II

Electric motor - An electric motor works by converting electrical energy into mechanical energy, using the principle that a magnetic field exerts a force on a current-carrying wire, causing it to rotate. This force generates torque that rotates a part of the motor, called the rotor or armature, within a stationary magnetic field produced by the stator. A mechanism called the commutator reverses the current direction in the armature at the right moment, allowing the rotation to continue in the same direction.

Stator - The stationary part of the motor that creates the magnetic field. This field can be generated by permanent magnets or electromagnets.

Rotor/Armature - The rotating part of the motor that contains coils of wire or can be a magnet.

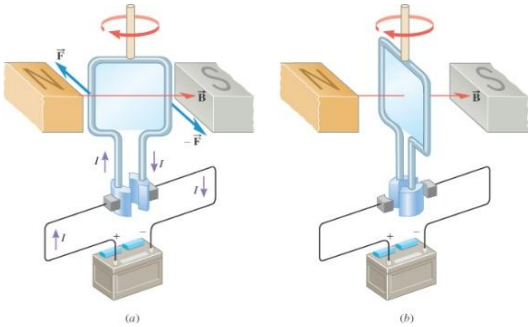
Magnetic Field & Force - When an electric current flows through the wires of the armature, it generates its own magnetic field, interacting with the stator's magnetic field. This interaction creates a force on the armature, causing it to spin.



Commutator (for DC motors) - As the armature spins, the commutator, a split ring connected to the armature, reverses the direction of the current in the coils at the appropriate times. This reversal ensures that the force continues to push the armature in the same direction, maintaining continuous rotation.

Brushes provide electrical contact between the stationary circuit and the rotating commutator, allowing current to flow into the armature.

dc Electric Motor - The diagram below shows the torque applied to the armature (wire loops) is produced by the current in the wire.

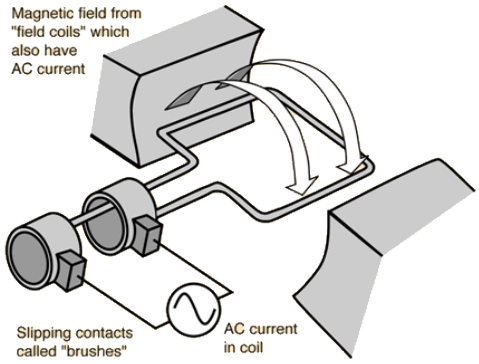


As the armature rotates through the magnetic field the conducting brushes keep electrical contact until the normal to the loop is along the field lines (part b) there is no torque but the momentum of the rotation causes the loop to overshoot the normal position and at the base the

brushes come in contact with the wire again but now the bushings contact the wire to product a current in the opposite direction, producing the torque in the same direction as before, this process is repeated as the armature spins between the magnets.

- As the armature rotates, the commutator switches the direction of the current in the armature's coils. This action reverses the polarity of the armature's electromagnet, ensuring that like poles always repel and unlike poles always attract, thus continuing the rotation.

ac Electric Motor - Alternating current is supplied to the stator's coils. Because the current constantly reverses direction, it creates a current in the coil that appears to rotate around the inside of the motor housing. This alternating current creates the force that rotates the coil.



Back emf in the electric motor – When an electric motor runs, its rotor (armature) spins inside a magnetic field. This means the conducting coil in the rotor are seeing a changing magnetic flux. According to Faraday's Law, a wire loop experiencing a changing magnetic flux experiences an induced voltage or emf. This induced voltage (V_{Bemf}) opposes the applied voltage that is driving the motor. This creates an emf that acts against the source voltage (V_{ext}) and is called "back" emf and acts to reduce the current flow in the motor.

There are two voltages involved in a motor.

1) Voltage applied externally to the motor (V_{ext}) from the power source.

2) Back (induced) emf due to rotating coils in magnetic field (V_{Bemf}) which has the opposite sign as the V_{ext}.

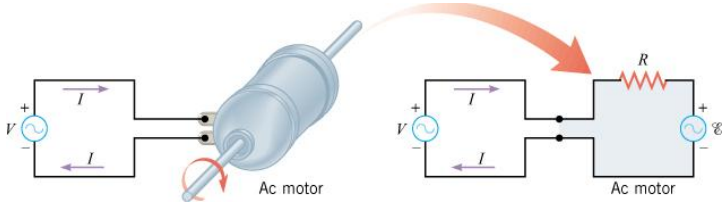
- The net voltage V_{net} is equal to the external voltage minus the induced voltage and defines how much current flows through the coils of the motor.

- The greater the speed of the motor, the greater the changing magnetic flux and the greater the back emf created.
- When the motor first starts up there is no Back V_{emf} and the current is max ($I = V_{ext}/r$).

Total current (I) drawn by motor – As soon as the motor is turned on, there is no back emf created yet and the current is maximum (V_{Bemf} = 0 V). As the motor's armature starts to turn the back emf begins to be created, this reduces the current in the circuit. When the motor reaches a steady state the back emf is at a max and the current drawn by the motor is at a minimum.

$$V_{net} = Ir_{int} = V_{ext} - V_{Bemf}$$

$$I = \frac{(V_{ext} - V_{Bemf})}{r_{int}}$$



- Where Ir_{int} is the voltage drop across the internal resistance, V_{ext} is the applied external voltage, V_{Bemf} is the back emf and r_{int} is the resistance of the wire in the coil and I is the current through the motor and the current supplied by the power source.

Example: A dc motor operates with a 12 V power supply and has an internal resistance of 6 Ω.

A) Determine the amount of current supplied to the motor at the start.

V_{Bemf} = 0 V at start up

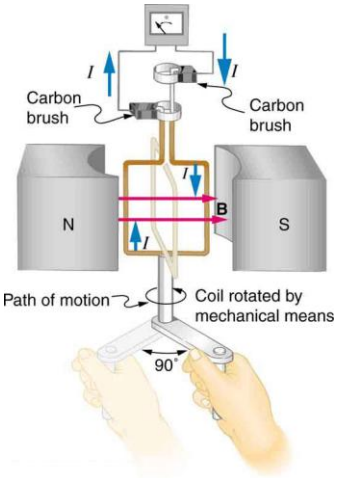
$$I = \frac{(V_{ext} - 0)}{r_{int}} = 2\ A$$

B) Determine the amount of current supplied to the motor after steady state rotation if the back emf generated is 10 V.

$$I = \frac{(12\ V - 10\ V)}{6\Omega} = 0.33\ A$$

Electric Generators – An a.c. Generator produces electrical energy from mechanical work by rotating a conducting wire coil (the mechanical work) in a magnetic field, the change in magnetic flux in the coil of wire induces an emf that produces a current in the coil.

Armature – coil of wire which the current is induced. A ferromagnetic core is usually placed inside the coil to increase the magnetic field strength thereby increasing the induced voltage.



There are two types of generators.

1) Rotating coil – in a generator when the wire coils are rotated and the magnetic is fixed.

2) Rotating magnet – when the magnetic is rotated and the coil of wire is fixed.

- The current varies sinusoidal as the coil is rotated through the field.
- This is the method used to produce most of the electrical energy by Electric Companies and is how the electrical energy is delivered to the home.
- Work must be done on the generator to keep the shaft moving against the counter-torque.

For both types of generators, the output voltage is the same and is given by

$V_{emf} = NAB\omega\sin(2\pi f\, t) = V_p\sin(2\pi f\, t)$ where $\omega = 2\pi f$

N – number of coils
A – cross-sectional area of coil
B – magnitude of magnetic field
ω – $2\pi f$ - angular velocity
V_p – peak voltage
T – period = 1/f

The value of the peak voltage is when $\sin(2\pi ft) = 1$ and is given by

$$V_p = NAB\omega = NAB2\pi f$$

The rms value is given by

$$V_{rms} = \frac{V_p}{\sqrt{2}} = \frac{NAB(2\pi f)}{\sqrt{2}}$$

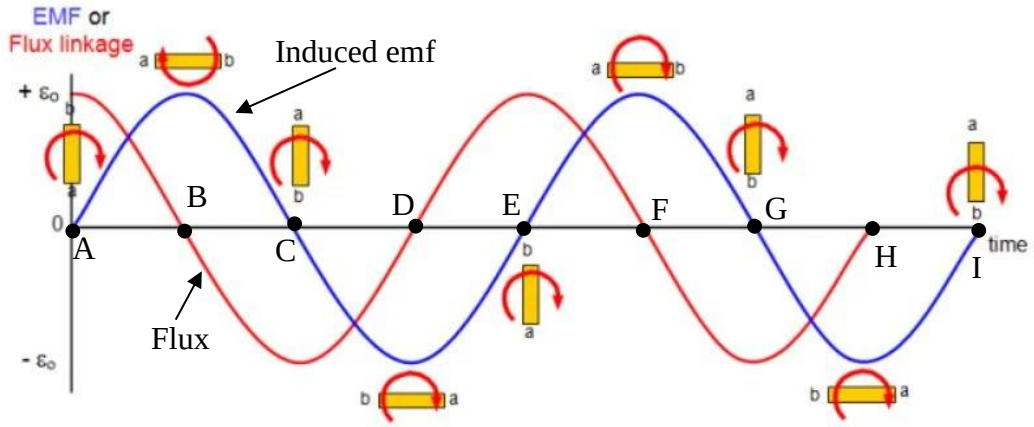
As a coil rotates within a magnetic field, the changing magnetic flux through the coil induces a voltage. This rotation causes the conductor to cut through the magnetic field lines at different angles, leading to a continuously changing induced voltage and current that alternates in direction, creating an AC output.

$$V_{emf} = -N\frac{d\Phi_M}{dt}$$

The induced voltage is proportional to the rate of change in magnetic flux. When the coil's orientation relates to the magnetic field changes, the direction of the induced current also reverses. This continuous rotation and changing flux results in a voltage that alternates in polarity, producing an alternating current.

- Maximum change in the magnetic flux creates the maximum induced voltage.
- The current varies sinusoidal as the coil is rotated through the field.
- Work must be done on the generator to keep the shaft moving against the counter-torque.

$$V_{emf} = -\frac{d\Phi_M}{dt}$$



The graph above shows the magnetic flux thorugh the generator coil (red line that is maximum at t = 0s) and the induced emf (blue line that is zero at t = 0s)

A - The induced voltage is given by the time rate of change of the magnetic flux. At t = 0 s the slope of the flux line is zero therefore the induced voltage is zero.

B - The maxium slope of the flux line gives the maxium induced voltage, the slope is negative therefore the induced voltage is postive.

C – The slope of the flux tangent line is zero therefore the induced voltage is zero.

D – The slope of the flux tangent line is maximum and is postive therefore the induced voltage is negative.

E - The slope of the tangent line for the flux is zero therefore the induced voltage is zero.

F – I the cycle repeats.

Example: Bob wants to create an electrical generator using the Earth's magnetic field. Determine the number of loops of wire he will need in the generator's armature if the magnetic field strength is $|\vec{B}|$ = 3 x 10^{-5} T, the area of the armature A = 0.2 m^2, the rms volage needs to produce is 120 V at 60 Hz.

Solve for the number of loops N

$$V_{rms} = \frac{NAB(2\pi f)}{\sqrt{2}}$$

$$N = \frac{V_{rms}\sqrt{2}}{AB(2\pi f)} = 75{,}000\ loops$$

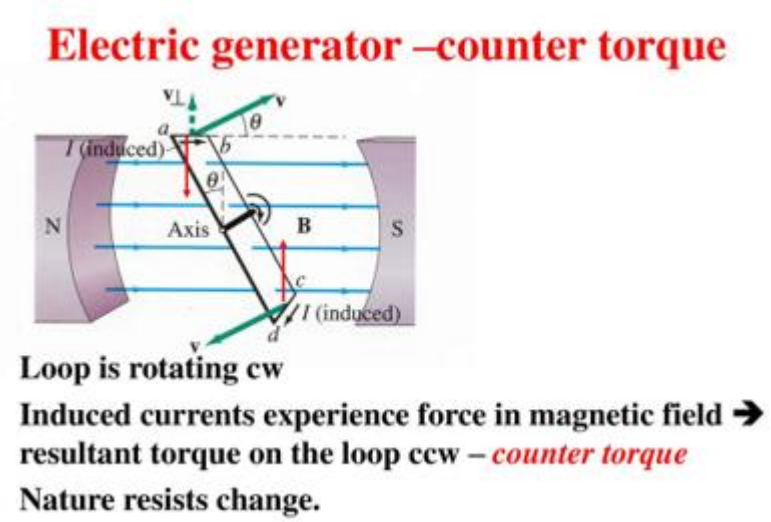
Counter Torque – is the torque created by the induced current in the coil, this induced current creates a magnet force due to the external field of the generator and opposes the motion of the coil. Because this torque is created by the induced current it only exists when there is an induced current if there is no current there is no counter torque.

- This is a real force that work must be done against to produce electrical energy.
- The more current generated the larger the counter torque.

Generator – by doing work on the shaft to rotate a coil in a magnetic field there is induced in the coil an electrical emf. (Current output)

Open circuit - no current so no force resisting rotation except friction

With load - current causes counter torque, a force that oppose the rotation

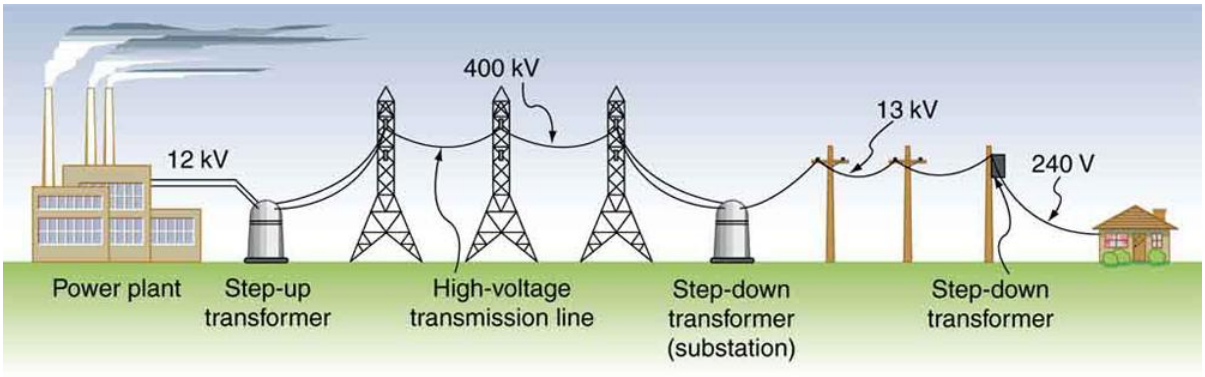


Difference between Motors and Generators

→ Motor uses current (Electrical energy) to rotate shaft (Mechanical energy) to perform work

→ Generator uses rotating shaft (Mechanical energy) to produce current (electrical energy)

Power Plants and transmission of electrical Power – Electrical power plants generate electrical power using generators that are driven by many different sources. Most Power plants uses high pressure steam to turn the generators, this steam is created by either burning some type of fuel (coal, natural gas, etc. …) or by nuclear reactions (fission reactions). Other generators use water behind a dam to turn the generator (hydroelectric) or wind that turns propellers to create electrical energy.



The power plant generates very high voltages (up to 12,000 V), using step up transfromers, this potenial is incresed to 400,000 V to transmit the power over long distance transmsion lines. This is done because the tranmission of voltage is more effiecnent than current. This Power is delieverd to substations that use step down transformers to deleiver the power to neighborhoods where another step down transform on the power poles is used to reduce the voltage even more to deviever the power to our house (220 V).

High Voltage Transmission of Electrical Power - Why do power plants use such High voltages (200,000 V to 800,000 V) to transmit electrical power? Because the electrical energy loss is less for higher voltages, it is the flow of charge (current) that creates the most loss of energy so the less current in the lines the better. Power lost by circuit $P_{loss} = I^2R$ and the Power Needed by circuit is $P_{need} = IV$

It is the same current (I) so

$$\mathbf{P_{loss}} = \frac{\mathbf{(P_{need})^2 R}}{\mathbf{V^2}}$$

Where R is the resistance in the power lines and can be considered constant.

- ✓ The larger the voltage V, the smaller the power loss by the square of V

Example: How much power is lost if the transmission voltage is changed from V_1 = 100,000 V to V_2 = 100 V? Assume the power needed and the resistance of the wires stay the same. Solve above equation for (P_{need}) R because they are the same for both.

$$\mathbf{(P_{need})^2 R} = \mathbf{[P_{loss\ 1}]}\boldsymbol{V_1^2} = \mathbf{[P_{loss\ 2}]}\boldsymbol{V_2^2}$$

$$\mathbf{P_{loss\ 1}} = \frac{\boldsymbol{V_2^2}\mathbf{[P_{loss\ 2}]}}{\boldsymbol{V_1^2}} = \mathbf{(1,000,000)P_{loss\ 2}}$$

There are two ways to reduce the loss in electrical transmission.

1) Use higher voltages - higher the voltage the lower the power loss throughout the circuit by $1/V^2$.
2) Use larger diameter cables – larger diameter cables reduce the resistances in the lines therefore reducing the resistive loss of the lines.

a.c. Power Transmission – The most common way power plants transmit their electrical energy to the customer is by using alternating current or voltage. Power plants use transformers to step up the generated voltage to transmit the power and then use step down transformers at the homes to step down the voltage before it reaches the homes.

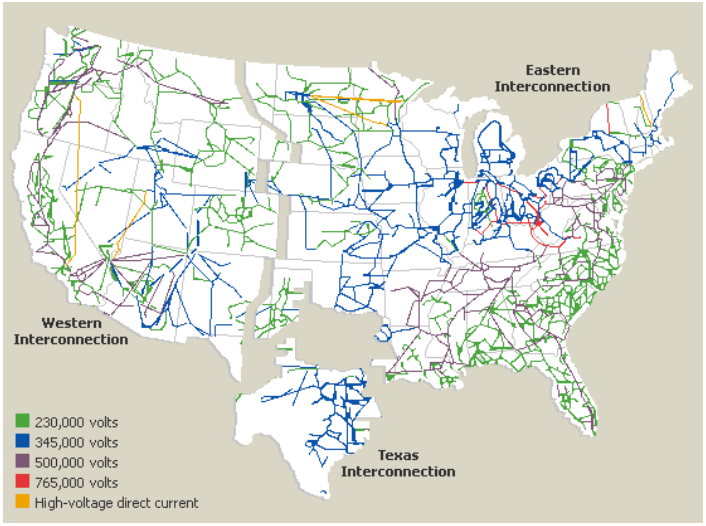
d.c. Power transmission (HVDC) – although most of the power transmission in the world uses a.c. voltage, there are advantages to d.c. voltage transmission. Direct current can generate about twice the voltages as that of a.c. thereby producing less loss in the transmission but transformers are more costly making it commonly used today for long distances or underground or under water transmission where a.c. transmission is more costly.

The Grid in the USA – is an interconnection of electrical power lines that connect the Electrical Generating Stations and suppliers to the customers (Businesses, Industry and households) . It is made up of high voltage transmission lines that transmit the power over long distances and distribution lines that transmit the electrical power to the customer. The Grid links major cities and towns to the power suppliers cross the region.

The Grid in the US is divided into three main geographic regions, the Eastern Interconnection, the Western Interconnection and Texas Interconnection. These main systems have limited connection with each other.

- The Grid can be affected by major power outages within the Grid, solar flares and heavy loads on the Grid.

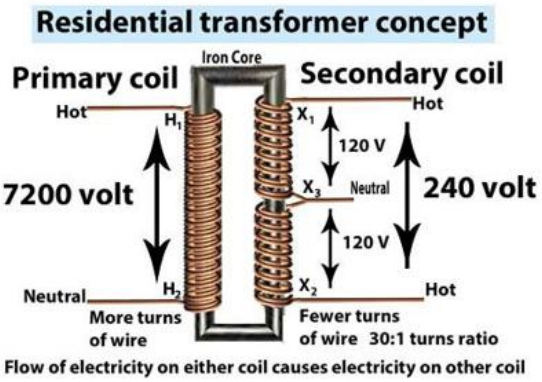


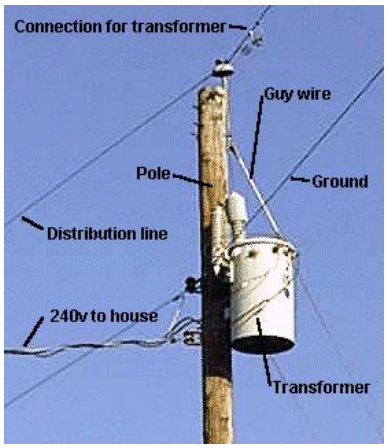


- Anybody can put energy back on the grid and the power companies must buy it back.

Electrical energy is distributed to a home from power plants via the electrical grid, which steps up high-voltage electricity for long-distance transmission, then steps it down at substations and neighborhood transformers to a safe, usable voltage for distribution to the home. Inside the home, electricity passes through a meter to a service panel where circuit breakers protect the circuits, then travels through wires in the walls to outlets and switches that power appliances.

240 volt electricity is created in a transformer by using a center-tapped secondary coil, which splits the 240 volt potential into two 120 volt lines that are 180 degrees out of phase. This "split-phase" system allows for both 120 volt circuits (using one hot and the neutral) and 240 volt circuits (using both hot wires from opposite ends of the coil) from a single source, with the center tap providing the neutral connection.





- ❖ Shown is the transformer (gray cylinder attached to pole) that converts the a.c. high volage on the distribution line to the a.c. 240 V that is delivered to the house.

Voltage in the Home - U.S. homes typically receive split-phase 240V service:

- Two 120V lines and a neutral.
- Most outlets use $120V_{rms}$.
- Large appliances (dryer, oven, HVAC) use 240V.

Service Drop - Overhead or underground line connects to your house.

Electric Meter - Measures how much electrical energy used (in kilowatt-hours).

Main Panel (Breaker Box) - Electricity enters your home through the main electrical panel. This panel contains circuit breakers that: The service panel contains circuit breakers or fuses that protect wires from overload by shutting off power to a specific circuit if it draws too much current. The main panel divides electricity into individual circuits. Common breakers are 10 A but larger appliances such as Air handlers and ovens are 20/40 A.

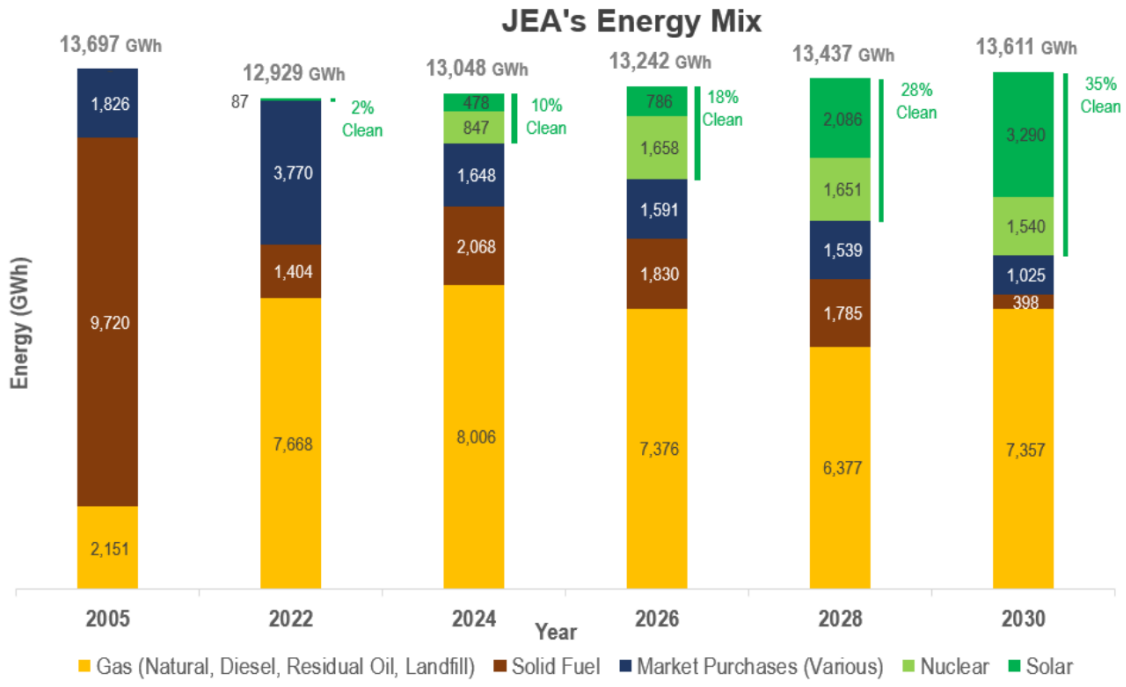
<u>JEA – Jacksonville Electric Authority</u> – produces the electricity that is used in the Jacksonville area, they have three main power plants that produce about 1,500 Megawatts of electrical power using coal, natural gas burning power generators and nuclear.



The image shows JEA's North Side power Plant, the tallest single tower is the exhaust tower from the burning of the coal.

The 2 shorter towers that are exhausting steam are the cooling towers, this is where the water is cooled off after it has moved through the generators, and the water is still very hot and must be cooled before it can be released back into the environment.

Energy Mix for JEA (taken from JEA.com) - JEA continues to diversify its electric generation portfolio with the addition of renewable energy resources. JEA's aim is to have up to 30 percent of energy provided to JEA's customers from carbon-neutral sources by 2030. The table below outlines the mix of fuels used to deliver reliable electricity to our customers.



<u>European power generation</u>

Standard Voltage and Frequency

- Most European countries use 230 V (single-phase) for residential and commercial supply.
- Frequency: 50 Hz is the standard across Europe.
- Three-phase systems: Typically, 400 V

Why 230 V and 50 Hz?

- Higher voltage (230 V) reduces current for the same power, minimizing energy loss and allowing thinner, cheaper wiring.
- 50 Hz was historically chosen in Europe for efficiency in early generators and transformers, and it remains the standard.

Europe's Electrical Grid Characteristics

- Europe's grid is AC-based, highly interconnected across countries for stability and energy sharing.
- Transmission: High-voltage AC lines (220 kV and above) for long-distance transport.
- Distribution: Lower-voltage AC for homes and businesses.
- Increasing integration of HVDC links for cross-border and offshore renewable connections, but the backbone remains AC.

Pros and Cons for USA and Europe Power Transmission.

Europe (230 V):

- Lower current → smaller wires → cost savings.
- Higher shock risk compared to 120 V.

U.S.A. (120 V):

- Safer for human contact.
- Higher current → thicker wires → more copper cost.